

Lecture 9: Volume scattering

Physical principles of remote sensing
(Ge 158)
Fall 2026
Brent Minchew

Remote sensing image of the day



SAR backscatter image of Sundarban National Park in India. “The Mangrove forest of the Sundarbans (right hand side of the river) is starkly contrasted with the harvested land on the left hand side of the river. The remains of the original Mangrove forest can be seen along the banks of the river at the ends of the strips of plowed fields. A number of boats are visible along the river.”

source: <https://www.capellaspace.com/americas-first-commercial-sar/>

Topics covered (see handwritten notes on Stellar)

- Fundamentals of volume scattering
- Heuristic model of scattering from vegetation
 - Principal types of scattering expected from a volume
 - Some common simplifications
- Heuristic model of scattering from a snow layer
- (Further reading, see Ulaby and Long (2014), Ch. 11)

Practical applications of today's lecture



Polarimetric SAR images of San Francisco. Red: $|HH+VV|$, green: $|HV|$, blue: $|HH-VV|$

Lee and Pottier, 2009



(A) Classification map

